

System Architecture & Design

Technical Reference Specification

Sample publication demonstrating core markpublish features: Typst typesetting, mathematical formulas, syntax highlighting, and callout boxes.

Version	2.1.10
Date	2026-09-23
Author	Frank Winter
Copyright	© 2026 Frank Winter

1 Introduction & System Architecture

markpublish transforms structured Markdown files into publication-grade documents. Structure, metadata, and styling are governed centrally in `markpublish.yaml` — everything else remains clean prose.

Note

The native Typst typesetting engine delivers book-grade printing, automatic hyphenation, and consistent typography without external LaTeX or Pandoc chains.

1.1 Modular Structure

Documents can be organized into parts, chapters, and divider pages. Each file remains pure Markdown, while the central manifest handles chapter ordering and automatic numbering.

2 Scientific Computations

Mathematical formulas are typeset with high typographic precision using the native Typst math engine:

$$E = \frac{mc^2}{\sqrt{1 - \frac{v^2}{c^2}}}$$

2.1 Pipeline Implementation

Source code blocks benefit from syntax highlighting with bundled fonts:

```
# pipeline.py
from pathlib import Path
import typst

def render_pdf(manifest_file: Path) -> bytes:
    """Compiles the document via Typst AST."""
    doc = load_manifest(manifest_file)
    return typst.compile(doc.to_typst())
```

3 Notes & Checklist

Critical information stands out with GitHub-style callouts:

Tip

A unified numbering pattern like 1| .1|+ synchronizes headings and tables of contents automatically.

Warning

Custom themes should always be validated locally using `markpublish build` prior to release.

3.1 Publication Checklist

- ☒ Validate project manifest `markpublish.yaml`
- ☒ Compile via Typst engine
- ☒ Verify accessible PDF standards
- ☐ Sign off publication